GERT VERCLEYEN

Curriculum Vitae

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Webpage



Employment

2024–present **Golomb Visiting Assistant Professor**, *Purdue University*.

Education

2019–2024 **PhD, Theoretical Physics**, *Maynooth University*, Ireland, Advisor: J. K. Slingerland.
On Low-Rank Multiplicity-Free Fusion Categories

2018–2019 **PhD, Theoretical Physics**, *Ghent University*, Belgium, Advisor: F. Verstraete.

2016–2018 **Master in Mathematics, Major in Theoretical Physics, Minor in research**, *Ghent University*, Belgium.

Research Interests

My research focuses on finding and probing properties of (multiplicity-free) fusion categories. To do so I write software using computational algebra to solve consistency equations arising from commutative diagrams. By developing software such as Anyonica.wl and FusionRings I managed to construct the largest database of known fusion rings and fusion categories, the AnyonWikiDatabase, which powers a collaborative platform, AnyonWiki that allows researchers to look up, download, and add data on Fusion Categories. Data from the AnyonWikiDatabase has already been incorporated in other packages such as TensorCategories.jl and TensorKit.jl.

Papers

2026 J. K. Slingerland and G. Vercleyen. On low-rank multiplicity-free complex fusion categories. (Accepted by Communications in Mathematical Physics), 2026.

2026 Fabian Mäurer, Ulrich Thiel, and Gert Vercleyen. F-symbols and R-symbols for the drinfeld center of the haagerup subfactor, 2026. arXiv:2601.20012.

2026 Colleen Delaney, Ralph Razzouk, Joost Slingerland, and Gert Vercleyen. Anyonwiki.org, 10 2026. arXiv:2610.30081.

2025 G. Vercleyen. Tables of practical invariants for distinguishing multiplicity-free fusion categories up to rank 7. 2025. arXiv:2507.00652.

2025 T. Maciazek, M. Conlon, G. Vercleyen, and J. K. Slingerland. Extending the planar theory of anyons to quantum wire networks. *SciPost Phys.*, volume 18, page 074. SciPost, 02 2025.

2025 Zhengwei Liu, Sebastien Palcoux, Yunxiang Ren, and Gert Vercleyen. Triangular prism equations and categorification, 2025. arXiv:2203.06522.

2024 G. Vercleyen. On low-rank multiplicity-free fusion categories. *PQDT - Global*, page 215, 2024.

2023 G. Vercleyen and J. K. Slingerland. On low rank fusion rings. *Journal of Mathematical Physics*, volume 64, page 091703, 09 2023.

Talks

The talks with a * are invited talks.

2026 **989 Fusion Categories Walk Into a Search Bar**, *Quantum Theory Neighborhood Seminar at Purdue University*.

2026 **Minimal fields and the center of the Haagerup fusion category**, *QuaSY-Con IV*.

2026 ***Complete invariants and minimal fields of multiplicity-free fusion categories**, *OSU Quantum Symmetry Seminar*.

2025 ***Complete invariants and minimal fields of multiplicity-free fusion categories**, *AMS Fall Sectional meeting*.

***The AnyonWiki, a digital repository for fusion categories and anyon models**, *AMS Fall Sectional meeting*.

***Techniques for categorifying multiplicity-free fusion rings**, *Mathematics seminar Kaiserslautern*.

Anyonica and the AnyonWiki, *Hopf25 Conference on Hopf algebras, quantum groups, monoidal categories and related structures*.

2024 **Anyonica and the AnyonWiki**, *ARTIN in Leeds, Conference on Biracks and Biquandles: Theory, applications, and new perspectives*.

***Knots and quantum computing**, *Topology Seminar Purdue*.

***Intro to Anyonica**, *Quantum Research Group Seminar at Purdue*.

Anyonica and the AnyonWiki, *Thematic Program in Field Theory and Topology at Notre Dame University*.

Anyonica and the AnyonWiki, *Great Lakes Mathematical Physics Meeting*.

***On Low-Rank Multiplicity-Free Fusions Categories**, *Quantum groups and monoidal categories seminar at Universite Libre Bruxelles*.

2022 ***Knots and quantum computing**, *Mathematics seminar at Osnabruck*.

***Knots and quantum computing**, *Doctoral mini-course on Combinatorial p-Kazhdan-Lustzig Theory*.

***Geometric Algebra**, *Maynooth theoretical physics seminar*.

Teaching

2024–2026 **Lecturer**, *Lectures on Ordinary Differential Equations (MA266) and Linear Algebra and Differential Equations (MA262)*, Purdue University.

2021 **Teaching Assistant**, *Tutorials on Special Relativity for 3rd-year physics students*, Maynooth University.

2021 **Teaching Assistant**, *Tutorials on Probability and Statistics for 3rd-year engineering students*, Maynooth University.

2021 **Teaching Assistant**, *Tutorials on Quantum Mechanics for 2nd-year physics students*, Maynooth University.

2020 **Teaching Assistant**, *Tutorials on Vectors and Matrices for 1st-year engineering students*, Maynooth University.

2019 **Teaching Assistant**, *Tutorials on Quantum Computing for graduate physics and mathematics students*, Ghent University.

Referees

Colleen Delaney
Assistant Professor,
Department of Mathematics,
Department of Physics and Astronomy
Purdue University
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Eric Rowell
Professor, Presidential Impact Fellow,
Department of Mathematics,
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Eddy Ardonne
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Assistant Professor,
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Steven Simon
Professor of Physics and
Professorial Fellow of Somerville College,
Rudolf Peierls Centre for Theoretical Physics,
Oxford University
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